

ALI NIKKHAH

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SUMMARY

ML Platform Engineer with 2+ years of experience building scalable ML infrastructure, LLM serving stacks, and MLOps platforms. At Digikala, engineered the production LLM serving layer handling 5M+ daily queries: vLLM with PagedAttention, Triton Inference Server, multi-tier Redis caching, async FastAPI, and multi-tenant Kubernetes deployment (200 pods, 40% GPU cost reduction). Designed the Modular Control Pipeline (MCP) for zero-downtime model hot-swapping and live index migrations. At HomaCloud, founded the ML platform from scratch: Kubernetes + Ray for distributed training, MLflow for experiment tracking, TensorFlow Serving for inference, ArgoCD/GitHub Actions for CI/CD (10min releases), achieving 15+ models in production with 99.9% deploy success rate. At Advanced Analytics Australia, built the edge ML serving infrastructure: TensorRT/TensorFlow Lite compilation, Triton serving, Kafka streaming for 100+ camera feeds. At Turquoise Digital, architected the enterprise data lake (Trino, PeerDB CDC 60s latency, 5TB/day) and deployed ML models for credit risk and portfolio allocation. Deep expertise in: GPU scheduling (MIG, NVIDIA device plugin), model optimization (quantization, TensorRT, ONNX), distributed training (Ray, PyTorch Distributed), high-concurrency serving (vLLM, Triton, TensorRT-LLM), and platform automation (Terraform, ArgoCD, Prometheus/Grafana).

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

ML Platform Engineer (LLM Serving)

Jan 2025 – Aug 2025

- Engineered the production LLM serving layer handling over **5 million daily queries**: integrated **vLLM** with PagedAttention and **Triton Inference Server** for LLM and embedding model deployment, maximizing KV-cache utilization and GPU throughput under variable query burst patterns (**3.2x throughput gain, 60% cost reduction per 1K queries, p99 200ms**).
- Implemented async **FastAPI** serving layers with multi-tier **Redis** caching and asynchronous request batching to fully isolate model execution latency from client-facing response times.
- Designed a Modular Control Pipeline (MCP) decoupling text ingestion, chunking, retrieval, and synthesis phases, enabling zero-downtime model hot-swapping and live index migrations without service interruption.
- Built stateful multi-agent routing networks using LangGraph cyclic directed graphs to manage multi-turn session state, dynamic context evaluation, and automated query rewriting loops.
- Architected multi-tenant Kubernetes deployment (200 pods) with GPU scheduling via NVIDIA device plugin and MIG, isolating vision and language model compute nodes for independent horizontal scaling and fault containment (**40% GPU cost reduction**); implemented HPA/VPA for automatic scaling under burst traffic.
- Established ML platform observability: Prometheus metrics for GPU utilization, KV-cache hit rates, request latency distributions; Grafana dashboards for real-time serving health; automated alerts for SLO violations.

Advanced Analytics Australia

Remote Contract

ML Platform Engineer (Edge)

Sep 2025 – Apr 2026

- Built the edge ML serving infrastructure for industrial safety compliance across **12 sites**: compiled and quantized vision models into **TensorRT** and **TensorFlow Lite** runtimes; served them through **Triton Inference Server** with dynamic batching, eliminating Python-level inference overhead (**3.5x speedup vs Python, 50ms inference**).
- Architected a high-throughput spatial metadata ingestion broker using **Apache Kafka** to buffer concurrent high-frame-rate tracking streams from **100+ cameras (30 FPS each)** without back-pressure on inference workers.

- Implemented edge MLOps workflow: zero-downtime model updates across distributed sites via Triton model repository; Prometheus/Grafana monitoring for inference health, model drift detection, and hardware failures; automated rollback on anomaly detection.
- Designed sensor calibration routines and signal-to-noise monitoring to maintain stable detection thresholds across indoor and outdoor deployment sites (**-35% false positives**).

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer

Sep 2025 – Mar 2026

- Architected the central enterprise data lake and OLAP warehouse for a high-volume BNPL and stock-trading platform, routing transactional and user-interaction streams from PostgreSQL into **Trino** via **PeerDB** Change Data Capture (CDC) replication for continuous, low-latency synchronization (**¡60s CDC latency, 5TB/day**).
- Designed and maintained complex **Apache Airflow** DAGs orchestrating multi-stage data transformation pipelines, automated cold-data archiving (Iceberg/Parquet on object storage), and cross-system synchronization workflows.
- Built a real-time statistical inference engine running parallel A/B hypothesis tests on application features using mSPRT sequential testing (**50+ concurrent experiments**); automated model deployment with FastAPI inference APIs, model versioning, and canary rollout with automated rollback.
- Deployed ML models for BNPL credit risk (AUC 0.87, -22% default rate) and time-series transformer portfolio allocation (Sharpe +0.42); implemented model monitoring for data drift, prediction drift, and performance degradation.
- Configured **Prometheus** and **Grafana** monitoring dashboards for pipeline latency, job failures, and storage degradation; integrated **Great Expectations** for automated data quality gating at ingestion.

HomaCloud

Tehran, Iran

Founding MLOps Engineer

Aug 2021 – Feb 2022

- Founded the ML platform from scratch: provisioned Kubernetes cluster with Terraform; deployed Ray for distributed training workload scheduling, MLflow for experiment tracking and artifact lineage, TensorFlow Serving for model inference.
- Automated end-to-end CI/CD with Docker, ArgoCD, and GitHub Actions, reducing release cycles to under 10 minutes (**99.9% deploy success rate**); implemented GitOps workflow for model and infrastructure deployments.
- Built low-latency model inference APIs with FastAPI and TensorFlow Serving to handle concurrent traffic spikes; orchestrated distributed training workloads with Ray and MLflow (**15+ models, 50+ experiments tracked**).
- Established ML engineering best practices: model registry with versioning, automated model testing (unit, integration, A/B), canary deployment with automated rollback, lineage tracking, and cost optimization via spot instances and GPU scheduling.

SELECTED PROJECTS & RESEARCH

Enterprise LLM Serving Platform

Digikala

ML Platform Engineer

2024 – 2025

- Built production LLM serving stack at 5M+ daily queries: vLLM PagedAttention + Triton, async FastAPI, multi-tier Redis, request batching, LangGraph session management — all on multi-tenant K8s (200 pods, GPU scheduling via MIG, 40% GPU cost reduction).
- Designed Modular Control Pipeline (MCP) for zero-downtime model hot-swapping, live index migrations, and canary deployments; established SLOs (p99 ¡200ms, 99.9% uptime) with Prometheus/Grafana observability.

Edge ML Serving Infrastructure

Advanced Analytics Australia

ML Platform Engineer

2025 – 2026

- Deployed TensorRT/TF Lite models via Triton on AXIS edge hardware across 12 sites; Kafka streaming for 100+ cameras @ 30 FPS; dynamic batching for 3.5x speedup; ¡50ms inference, 99.2% uptime.
- Built edge MLOps: zero-downtime model updates via Triton repository, automated rollback on anomaly detection, Prometheus/Grafana monitoring for inference health, drift detection, hardware failures.

Enterprise Data Lake & ML Deployment Platform

Turquoise Digital

Senior Data Engineer

2025 – 2026

- Architected Trino + PeerDB CDC data lake (¡60s latency, 5TB/day); Airflow DAGs with Iceberg/Parquet; PySpark ETL (60% memory reduction); Great Expectations quality gates.
- Deployed ML model serving: FastAPI inference APIs, model versioning, A/B testing (mSPRT), canary rollout with automated rollback; model monitoring for data/prediction drift.

ML Platform Foundation

HomaCloud

Founding MLOps Engineer

2021 – 2022

- Built ML platform from scratch: K8s + Ray (distributed training) + MLflow (experiment tracking) + TF Serving; CI/CD (ArgoCD, GitHub Actions, Terraform ¡10min); GitOps for model/infra deployments.
- Established ML engineering practices: model registry, automated testing (unit/integration/A/B), canary deployment with rollback, lineage tracking, cost optimization via spot instances and GPU scheduling. 15+ models, 99.9% deploy success.

Distributed Training & Serving Optimization

Personal Research

ML Platform Engineer

2024

- Benchmarked LLM serving: vLLM PagedAttention vs TensorRT-LLM vs Triton; optimized batching, KV-cache, GPU utilization; published findings on serving 7B/70B models at scale.
- Explored distributed training: PyTorch Distributed, Ray Train, DeepSpeed; compared throughput, scaling efficiency, fault tolerance for multi-node LLM fine-tuning.

RESEARCH HIGHLIGHTS

University of British Columbia

Remote Research Collaboration

Research Assistant – Vision-Language Medical Imaging

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Remote Research Collaboration

Research Assistant – Video Motion Classification

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems *GPA: 3.65 overall, 3.91 major*

Sept 2020 – Jul 2024